William Timkey

Curriculum Vitae

Department of Linguistics
New York University
wpt2011@nyu.edu
wtimkey.github.io
Github *Linkedin*

Last Updated: October 2026

Education

2022–present **PhD, Linguistics (expected Dec. 2027)**, *New York University*, Advisor: Tal Linzen.
Core Dissertation Committee: Tal Linzen, Brian Dillon, Alec Marantz

2017–2020 : **Bachelor of Arts, Linguistics**, *Cornell University*, With Distinction, GPA: 4.075.
Transferred in from SUNY Buffalo

Other Education (Non-degree)

2021–2022 : **Korean Language Student**, *Seoul National University KLEC*.

2016–2017 : **Bachelor of Science, Computer Science**, *SUNY Buffalo*, Advanced Honors Program, GPA: 4.000.

Publications

Journal Articles (published)

2026 **Timkey, William**, Kuan-Jung Huang, Byung-Doh Oh, Grusha Prasad, Suhas Arehalli, Tal Linzen, and Brian Dillon. Eye movements reveal a dissociation between prediction and structural processing difficulty in language comprehension. *Proceedings of the National Academy of Sciences*, volume 123. Proceedings of the National Academy of Sciences, August 2026.

Journal Articles (in prep/review)

2026 **William Timkey**, Brian Dillon and Tal Linzen. Why are language models less surprised than humans? Testing the Parse Multiplicity Mismatch Hypothesis. In review, preprint available, May 2026.

2026 **William Timkey**, Brian Dillon and Tal Linzen. A rational neural network model of memory retrieval for syntactic dependency resolution. In prep, preprint available on request, October 2026.

Conference Proceedings

2023 **Timkey, William** and Tal Linzen. A language model with limited memory capacity captures interference in human sentence processing. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 8705–8720, Singapore, December 2023. Association for Computational Linguistics.

2021 Matt Wilber, **Timkey, William**, and Marten van Schijndel. To point or not to point: Understanding how abstractive summarizers paraphrase text. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3362–3376, Online, August 2021. Association for Computational Linguistics.

2021 **Timkey, William** and Marten van Schijndel. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 4527–4546, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics.

Conference Proceedings (in prep/review)

2026 **William Timkey** Brian Dillon and Tal Linzen. Parsing as an optimal transport problem: Accounting for syntactic disambiguation difficulty. In prep, preprint available on request, October 2026.

2026 **William Timkey** and Tal Linzen. Do syntax heads influence model behavior? In review, preprint available online, 2026.

Other Conference Presentations

2026 **William Timkey,** Brian Dillon, Tal Linzen, A Rational Neural Network Model of Memory Retrieval for Syntactic Dependency Resolution, In *Architectures and Mechanisms for Language Processing (AMLaP) 2026* (oral).

2026 **William Timkey,** Kuan-Jung Huang, Byung-Doh Oh, Grusha Prasad, Suhas Arehalli, Tal Linzen, Brian Dillon, Prediction and structural processing dissociate: A large-scale eyetracking study , In *The 39th Annual Conference on Human Sentence Processing*.

2025 **William Timkey** and Tal Linzen, Language model surprisal underpredicts garden path effects even with limited syntactic parallelism, In *The 38th Annual Conference on Human Sentence Processing*.

2024 **William Timkey**, Kuan-Jung Huang, Grusha Prasad, Suhas Arehalli, Tal Linzen, Brian Dillon, Large-scale eye-tracking while reading benchmark shows surprisal captures early fixations, but not regressions, In *The 37th Annual Conference on Human Sentence Processing*.

Invited Talks

2026 *Title TBD.* MIT Computational Psycholinguistics Lab, Massachusetts Institute of Technology

2026 *Using neural language models to test theories of human sentence processing.* Laboratoire de Sciences Cognitives et Psycholinguistique (LSCP), École Normale Supérieure

2025 *What a large-scale eye tracking benchmark reveals about syntactic reanalysis in reading.* Sentence Processing Workshop, University of Potsdam

2025 *Using Neural Networks to Model Syntactic Dependency Resolution Difficulty in Human Sentence Processing* Psycholinguistics group, University of Massachusetts, Amherst

2024 *What can LMs tell us about how humans resolve syntactic dependencies?* Interpretable, Cognitively inspired, Low-resource Language Models (InClow Lab), University of Groningen

2024 *Toward a rational model of working memory in parsing.* Language, Computation, and Cognition Lab (LaCoCo), Saarland University

2024 *Toward a rational model of cue-based retrieval* Joint UMass/UMD Psycholinguistics Workshop, University of Massachusetts, Amherst

Research Positions

Aug 2022 – Present **Computation and Psycholinguistics Lab, New York University**, PhD Student.
Advisor: Prof. Tal Linzen

Jan 2025 – Present **Computational Sentence Processing Lab, University of Massachusetts, Amherst**, Visiting PhD Student.
Advisor: Prof. Brian Dillon

Aug 2020 – Dec 2021 **Computational Psycholinguistics Research Discussions Group (C.Psyd), Cornell University**, Research assistant.
Advisor: Prof. Marten van Schijndel

Aug 2020 – Apr 2021 **Department of Linguistics, Cornell University**, Historian.
Advisor: Prof. John B. Whitman

Aug 2019 – Jan 2020 **Department of Information Science, Cornell University**, Independent Research.
Advisor: Prof. David Mimno

May 2019 – Jan 2020 **Human-Computer Interaction Institute, Carnegie Mellon University**, Research Intern.
Advisors: Prof. Brad Myers and Prof. Toby Li

May 2019 – Jan 2020 **Collaborative Technologies Lab, Cornell University**, Research Assistant.
Advisors: Prof. Susan Fussell

Service

Positions

2026- **Reviewer** *ACL Rolling Review (ARR)*

2023-2024 **Secretary** *The Linguistics Association of NYU (LANYU)*

2023 **Graduate School Application Panel** *Department of Linguistics, New York University*

2024 **Bibliography Workshop Organizer** *Department of Linguistics, New York University*

2020 **Volunteer** *Conference on Emperical Methods in Natural Language Processing*

Ad-hoc Journal Reviewing

2026- Open Mind

2026- Glossa Psycholinguistics

2025- Cognition

2024- Journal of Memory and Language

Undergraduate Advising

2026- Lukeria Petrunina, NYU Lingusitics

Teaching

Spring, 2026: **LING-UA 61: Human Sentence Processing**, *New York University*, Teaching Assistant.

Fall, 2024: **LING-UA 27: Language and Mind**, *New York University*, Teaching Assistant.

Fall, 2023: **LING-UA 27: Language and Mind**, *New York University*, Teaching Assistant.

Fall, 2021: **STAT 104: Introduction to Quantitative Methods for Economics**, *Harvard University*, Teaching Fellow.

Summer, 2021: **STAT 100: Introduction to Quantitative Methods**, *Harvard University*, Teaching Fellow.

Spring, 2021: **STAT 102: Introduction to Quantitative Methods for Life Sciences**, *Harvard University*, Teaching Fellow.

Fall, 2020: **STAT 104: Introduction to Quantitative Methods for Economics**, *Harvard University*, Teaching Fellow.

Summer, 2020: **STAT 100: Introduction to Quantitative Methods**, *Harvard University*, Teaching Fellow.

Spring, 2020: **INFO 2950: Introduction to Data Science**, *Cornell University*, Teaching Assistant.

Fall, 2017: **CS 199: Introductory Computer Science Seminar**, *SUNY Buffalo*, Teaching Assistant.

Awards

2026 NYU Faculty of Arts and Sciences Dean's Conference Fund Award Recipient

2025 NYU Faculty of Arts and Sciences Dean's Conference Fund Award Recipient

2022-present NYU Graduate School of Arts and Sciences Henry M. Mccracken Fellowship

2022 Seoul National University KLEC Academic Honor Scholarship

2019 Erasmus Mundus scholarship (declined due to COVID-19 pandemic travel restrictions)

2017 SUNY Buffalo Advanced Honors College

2017 SUNY Buffalo Provost Scholarship

Skills

Programming Languages: python (pytorch, pandas etc.), R (lme, brms, etc.), SQL, Java, HTML, CSS, javascript, PHP

Human Languages: **English** (native), **Korean** (high intermediate/advanced), **German** (low intermediate)